

# Fractions on the Number Line

## Answer Key

### Grade 3 — Fractions as Numbers

#### Quick Answers

---

- |                                         |                                               |
|-----------------------------------------|-----------------------------------------------|
| 1. $\frac{3}{4}$                        | 7. $\frac{5}{3}$                              |
| 2. $\frac{3}{4}$                        | 8. $>$                                        |
| 3. $\frac{3}{8}$                        | 9. (b) number of jumps from zero = 6, —       |
| 4. (b) number of jumps from zero = 5, — | 10. (a) how many eighths reach the same point |
| 5. (b) number of jumps from zero = 3, — | = 6, —                                        |
| 6. $<$                                  |                                               |
- 

#### What is verified

---

6 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 4, 5, 9, 10. The worked solution states what a correct response must contain.

---

#### Curriculum

---

**Level:** Grade 3 — Fractions as Numbers

3.NF.A — *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

*Difficulty 1–5 of 5 across 14 tagged checks.*

---

**1.** The whole from 0 to 1 is cut into 4 equal parts. What fraction does the marked point show?

**Solution:** One whole is cut into 4 equal parts, so one jump is  $\frac{1}{4}$ . Count jumps from 0 to the dot:  $\frac{1}{4}, \frac{2}{4}, \frac{3}{4}$  — three jumps. The point is three jumps of one fourth.

$\frac{3}{4}$

---

**2.** This whole is cut into 3 equal parts. What fraction does the marked point show?

**Solution:** Three equal parts in the whole, so one jump is  $\frac{1}{3}$ . The dot is on the second tick after 0, which is two jumps.

$\frac{2}{3}$

---

**3.** This whole is cut into 8 equal parts. What fraction does the marked point show?

**Solution:** Count the equal parts between 0 and 1: there are 8, so each jump is  $\frac{1}{8}$ . The dot sits on the fifth tick after 0. Counting matters more than measuring here — the eighths are narrow, so count ticks rather than guessing by eye.

$\frac{5}{8}$

---

4. A line cut into 6 equal parts. (a) Mark  $\frac{5}{6}$ . (b) How many jumps of  $\frac{1}{6}$  from 0 is that?

**Solution (a) — judged by eye.** The dot belongs on the fifth tick to the right of 0, which is one tick short of 1. A dot between two ticks, or on the sixth tick (which is 1), is not correct. The bottom number of the fraction tells you how many parts the whole was cut into; the top number tells you how many to count off.

**Solution (b):**  $\frac{5}{6}$  means five copies of  $\frac{1}{6}$ , so it is five jumps:  $\frac{5}{6} \div \frac{1}{6} = 5$ .

5

---

5. A line of halves running past one whole. (a) Mark  $\frac{3}{2}$ . (b) How many jumps of  $\frac{1}{2}$  from 0 is that?

**Solution (a) — judged by eye.** Each part is a half, so the ticks fall at 0, one half, 1, one and a half, 2. The dot belongs on the third tick after 0 — halfway between 1 and 2. This is the first fraction on the sheet bigger than one whole, and the number line makes that ordinary: keep jumping past 1.

**Solution (b):**  $\frac{3}{2}$  is three copies of  $\frac{1}{2}$ , so three jumps.

3

---

6. Two points on one line of eighths. Compare  $\frac{3}{8}$  and  $\frac{5}{8}$  with  $>$ ,  $<$ , or  $=$ .

**Solution:** On a number line, the number further to the right is the larger one. Both points are counted in the same jumps of  $\frac{1}{8}$ , so the one with more jumps is further right: five jumps beats three.

$$\frac{3}{8} < \frac{5}{8}$$


---

7. A line from 0 past 1 with every whole cut into 3 equal parts. What fraction does the marked point show?

**Solution:** Each jump is  $\frac{1}{3}$ . Keep counting past 1:  $\frac{1}{3}, \frac{2}{3}, \frac{3}{3}, \frac{4}{3}, \frac{5}{3}$ . The dot is on the fifth tick after 0, which is  $\frac{5}{3}$  — two thirds of the way from 1 to 2. A fraction may be bigger than one whole; the top number simply passes the bottom one.

$\frac{5}{3}$

---

8. The top line is cut into fourths and the bottom into sixths, with one point marked on each. Compare  $\frac{3}{4}$  and  $\frac{3}{6}$ .

**Solution:** Do not compare the top numbers — they are both 3, and that tells you nothing on its own. The lines are the same length, so read position: three fourths sits most of the way to 1, while three sixths sits exactly halfway. The point further right is larger.

$$\frac{3}{4} > \frac{3}{6}$$


---

9. A line cut into 8 equal parts. (a) Mark  $\frac{6}{8}$ . (b) How many jumps of  $\frac{1}{8}$  from 0 is that?

**Solution (a) — judged by eye.** The dot belongs on the sixth tick after 0 — two ticks short of 1. Counting from 1 backwards ( $\frac{7}{8}$ , then  $\frac{6}{8}$ ) is a fine strategy and usually faster than counting up.

**Solution (b):**  $\frac{6}{8}$  is six copies of  $\frac{1}{8}$ , so six jumps.

6

---

10. Two lines of the same length, cut into fourths and into eighths, with  $\frac{3}{4}$  marked on the top one. (a) How many eighths reach the same point? (b) Explain how the lines show it.

**Solution (a):** Ask how many  $\frac{1}{8}$  jumps fit into  $\frac{3}{4}$ :  $\frac{3}{4} \div \frac{1}{8} = 6$ . Reading it off the picture works too — every fourth-tick has an eighth-tick sitting on top of it, with one extra eighth-tick between. 6

**Solution (b) — instructor-judged.** A full-credit answer says the two dots land on exactly the same place on the line, and explains why: each fourth is covered by two eighths, so three fourths takes  $3 \times 2 = 6$  eighths. Model answer: “The lines are the same length. Every jump of a fourth is the same as two jumps of an eighth, so three fourths is six eighths, and both dots are in the same spot.” Repeating the number six with nothing about the lines lining up is not full credit.